

Machine Tool Technology
Week-12

- 1. Introduction to Drilling - Types of Drilling Machines-Specification of Drilling Machines-Specification of Drill Bit, Reamer, Die and Taps**
- 2. Nomenclature of Drill Bit, Reamer, Die and Taps- Standard sizes of Drill Bits Explain Operations performed in Drilling Machines- Drilling, Counter sinking, Reaming, Boring. Tapping**
- 3. Prepare the Process plan as per the given drawing and Estimate the Production cost per Piece as per the given drawing**

DRILLING MACHINE

A drilling machine is a type of machine in which the holes are being made on the workpiece by making use of a rotating tool called drill bit or the twist drill.

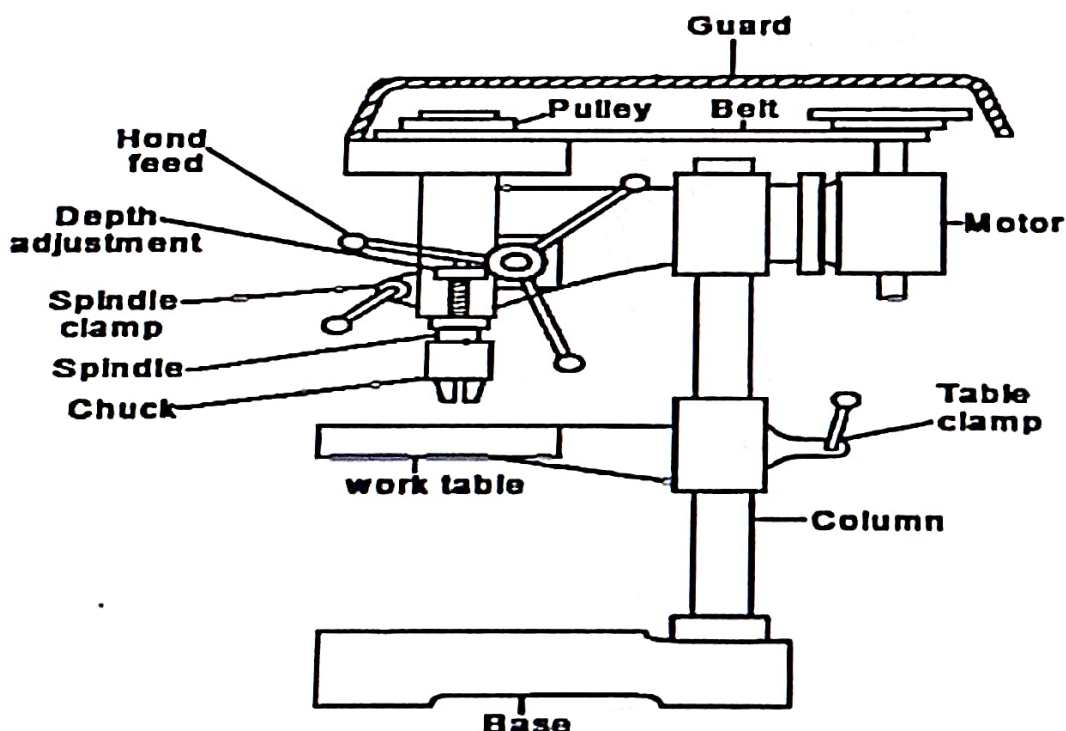
Drilling is basically a technology of creating holes. Drilling operations can also be performed on Lathe machines.

Drilling Machine has based upon the principle that the rotating edge of the tool exerts a large force on the workpiece and holes are being created in the workpiece. The material is removed from the workpiece by the shearing and extrusion process.

Drilling Machine Parts:

Drilling Machine consists of the following Main Parts:

- Bed
- Pillar
- Swivel table
- Motor
- Steeped pulley
- Spindle
- Chuck
- Drill Bit and
- Hand-wheel.



1. Bed:

The bed is the main part of the machine on which the whole machine is being mounted. The bed is made up of cast iron, so it has high compressive strength and good wear resistance.

2. Pillar:

The pillar is a type of vertical column that rests on the bed. A pillar is present at the center of the bed. The pillar helps the motor and the spindle head.

3. Swivel Table:

The table is the place where the workpiece is being mounted. The table is attached to the column and it can be rotated around the column and can have an upward and downward movement. A table can be adjusted at any angle as per the requirement.

4. Motor:

The motor is present at the top of the column. The inside motor shaft is there which is connected to a stepped pulley so that we can increase or decrease the speed of the rotation of the motor.

5. Stepped pulley:

Two stepped pulleys are present on either side of the column at the top. Out of these two, one pulley will be in an upward direction while the other pulley is inverted.

Always both the pulleys will be there in the opposite direction. The basic function of the stepped pulley is to control the speed of the rotation of the motor.

6. Spindle:

Spindle arrangement is present at top of the column opposite to the arrangement of the motor.

The top of the spindle is attached to one of the stepped pulleys. The bottom of the spindle is connected to the chuck.

7. Chuck:

Chuck is present at the bottom of the spindle. The basic function of the chuck is to hold the cutting tool firmly.

8. Drill bit:

A drill bit is an actual cutting tool that is used to create a hole in the workpiece.

9. Hand Wheel:

The basic function of the handwheel is to adjust the spindle position as per the requirement.

Drilling Machine Specification:

The following specification of Drilling Machine are:

- The portable Drill Machine is specified by the maximum diameter of the drill which can be held.
- Sensitive and the upright drill machine are specified by the diameter of the largest workpiece that can be held.
- The radial drill machine is specified by the length of the arm and column diameter.
- Multiple spindle drill machine is specified by the drilling Area, the size and the number of holes the machine can drill.

Drilling Machine Types:

The different types of Drilling Machine are:

- Portable Drilling Machine
- Sensitive Drilling Machine
- Upright or Column Drill Machine
- Radial Drilling Machine
- Gang Drilling
- Multi-Spindle Drilling
- Vertical Turret Type Drilling
- Automatic Drilling Machine
- Deep hole Drilling
- Turret Drilling
- Automatics Drilling Machine.

1. Portable Drilling Machine:

Portable Drilling Machine is the type of machine which consists of a small electric Motor which gives power for rotation of the drill bit. **Hand Drill** is the best example of the portable hand Drill.

Portable Machine is most helpful in the case where the job cannot be taken to the workshop due to their large size or weight so in such cases Portable drilling machines are best as they can be taken to the job(workpiece) and drilling can be done there itself.

Thus, it saves a lot of time and prevents transportation costs.

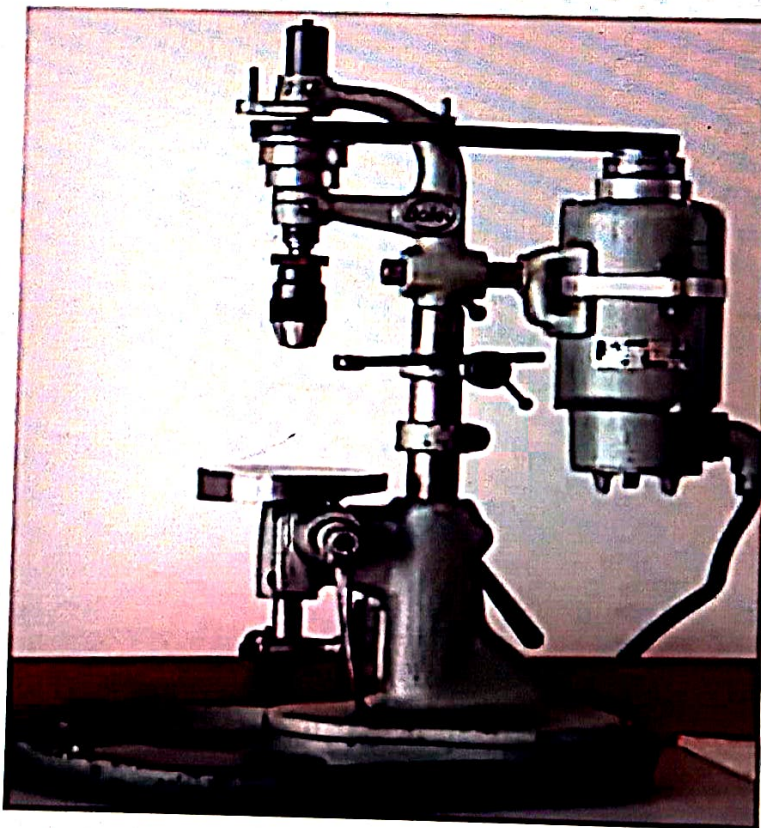


2. Sensitive Drilling Machine:

A sensitive Drilling machine is a machine that is hand-fed and operates on the principle of rack and pinion drive.

This is a light and simple bench-type machine used for light-duty work. In this, the drill head is counterbalanced and fed through a hand lever system.

The operator puts the force on the drill bit by manual rotation of the wheel, thus creating a hole in the workpiece.



3. Upright or Column Drilling Machine:

Upright or column drill is similar to the sensitive drill machine the only difference is that the power feeding mechanism is used for the rotation of the drill.

This machine is mostly suited for heavy work. The size of work that can be accommodated is limited by the distance between the spindle and column.

These machines mostly have a box-type column and are mostly used for heavier work.



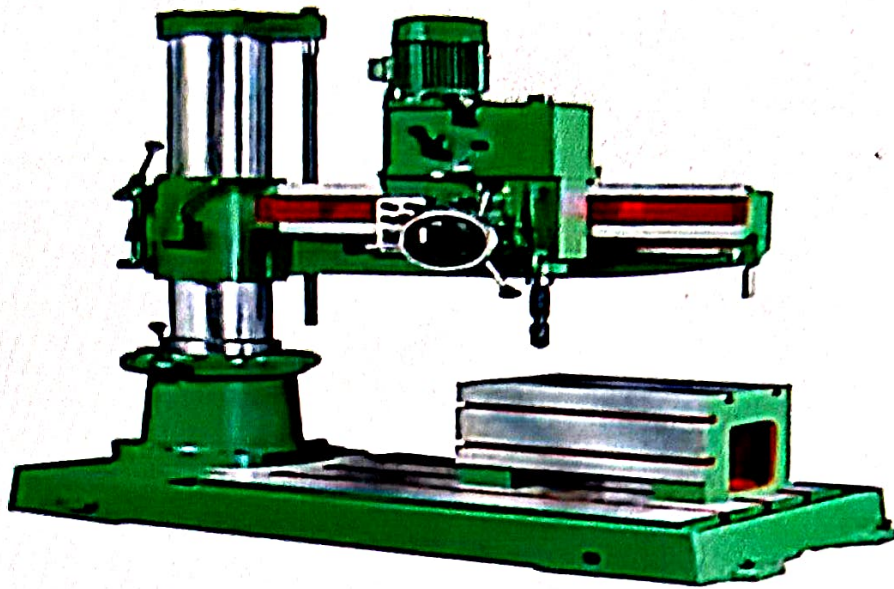
4. Radial Drilling Machine:

A radial drill machine is the most advanced form of the drill machine. It is the most versatile drill machine. Best suited for drilling a large number of holes.

The radial drill machine has a single-spindle machine intended for handling large and heavy work which is beyond the capacity of the simple machine.

It consists of a vertical column with a radial arm that can swing through an arc of 180 degrees or more.

The drilling head is equipped with a power feed. The drilling head may be moved along the arm by hand or power on gear and the track arrangement.



5. Gang Drilling Machine:

Gang Drilling Machines of two or more drill heads mounted on the same table. These can run simultaneously or in sequence. The number of spindles that can be mounted is unlimited but mostly four spindles are used.

It is most useful in the case when multiple operations are to be performed such as drilling, counterboring, reaming, etc. for drilling holes of several different sizes.

Application: Machines are used for the straight line, multiple hole drilling applications as in the pipe, channels, castings, plates, and angle.



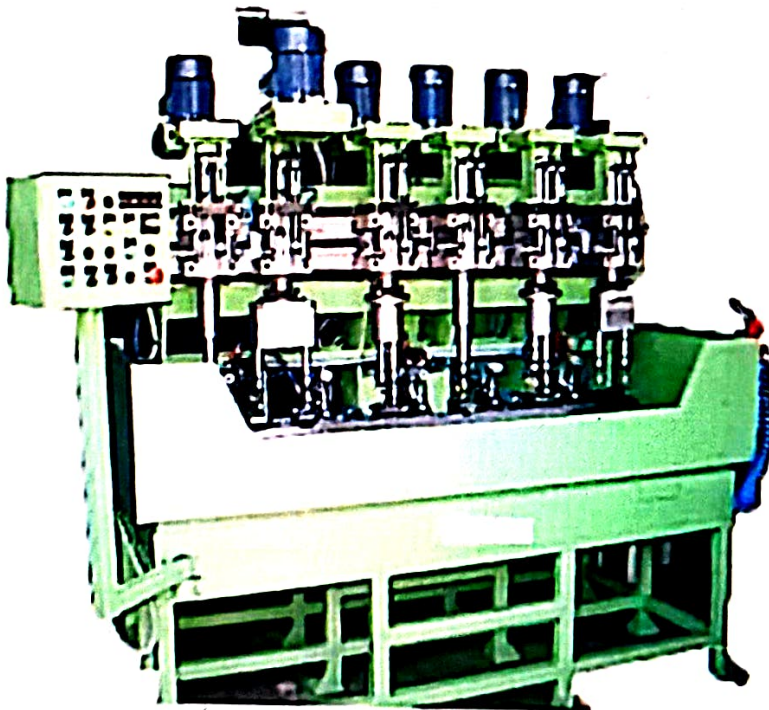
6. Multi-spindle Drilling Machine:

A multi-spindle drill Machine is developed for the purpose of drilling several holes simultaneously.

These machines are mostly used for larger production work where it is required to drill many parts with such accuracy that all the parts are interchangeable.

For drilling closely spaced holes, some of the holes are drilled first by a set of spindles then the job is repositioned and other sets of closely-spaced holes are then drilled by another set of spindles.

The special type of multi-spindle is the way-type. This type of machine is used in the case where several holes in the different planes are to be drilled.



7. Vertical turret type drilling Machine:

Vertical turret type drill machine consists of a turret that houses various tools such as drill, ream, spot-face, counterbore, tap in any desired sequence.

The various spindles on this turret can be indexed manually or automatically. The spindles drive only after they come to the drilling position.

8. Automatic Drilling Machine:

An automatic drill machine is used mostly in places of higher Production work. They consist of a number of unit heads with single or multiple spindles in angular, vertical or horizontal positions on various combinations on a special case.

Each station is also provided with an indexing table and work holding fixtures.

9. Deep-hole Drilling Machine:

Deep hole Drilling machines are used for drilling such holes whose length exceeds three times the drill size.

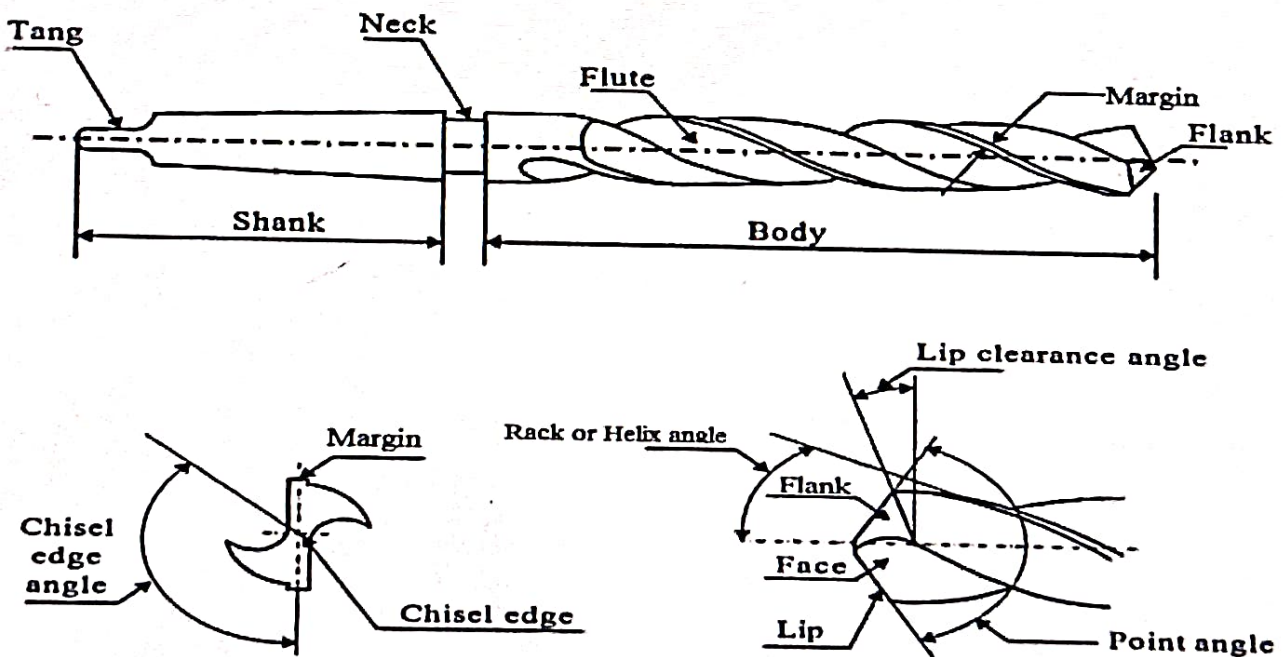
The special features of the Deep Drill Machine are:

- Robust design.
- Extreme shot setup time.
- Good quality of production.
- High-speed operation.
- Versatility: The various example of this class are riffle barrels, long Spindles, Connecting Rods, and certain oil well drilling equipment.

TWIST DRILL

A **twist drill** is basically a cylindrical piece of steel with special grooves. One end of the cylinder is pointed and the other end is so shaped that it can be attached to the drilling machine. The **grooves** are usually called **flutes**. The flutes formed by twisting a flat piece of steel into a cylindrical shape and such types of cylindrical shape drills are called **twist drills**.

Twist Drill Nomenclature:



The following Nomenclature of Twist Drill are:

- Body
- Shank
- Dead Center
- Point
- Lips
- Body clearance
- Chisel edge
- Chisel edge corner
- Face
- Flank
- Flutes
- Heel
- Neck
- Tang

Body: The portion between the shank and the drill bit tip is called 'Body'. The body is mostly fluted and relieved.

Shank: The part of the drill bit that holds into the holding is called the 'shank'.

Dead Centre: The sharp edge at the extreme tip end of the drill is formed by the intersection of the cone-shaped surfaces of the point. The dead center is always at the exact center of the axis of the drill.

Point: The entire cone-shaped surface at the cutting end of the tool.

Lips: The main cutting edges of the drill are formed by the intersection of the flank and the flute surfaces. For a good cutting, it should be straight, symmetrical with the axis of the shaft and equal in length.

Body clearance: To provide diameter clearance the body surface diameter is reduced.

Chisel edge: The chisel edge is the point. Here two cutting lips meet at extreme tip.

Chisel edge Angle: The chisel edge angle is the angle between the chisel edge and cutting lip measured plane normal to the axis.

Face: The flute surface portion adjacent to the lip, when it is cut from the work the chip impinges.

Flank: Drill surface which extends behind the lip to flute.

Flutes: The Twist Drill body has grooves and it is known as flutes.

Heel: The Heel is the intersection of the flute surface and the body clearance.

Neck: Neck is the portion of the body with reduced diameter between body and shank.

Tang: The tang is flattened end of the taper shank.

REAMING

Reaming is a finishing process that is performed with the multi-edged tools that provide high precision holes and also used for enlarging or finishing a hole previously drilled without changing the chemical and physical properties.

The reaming process took after the drilling process it removes a relatively small amount of material.

No specific machine is used for reaming can be performed on drilling machine, lathe machine, milling machine, and machining centres or by hands.

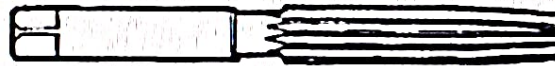
Purpose of reaming :

Reaming is applied to accurately finish drilled holes with a good surface finish and size. It offers the advantages that a greater number of holes can be produced with consistently good quality.

- Reaming required which has exact diameter required.
- It requires has an adequate edge profile.
- It is required where the well-sharpened surface needed.
- Reaming calls for a hole that has been exactly pre-machined.
- The use of lubricants will prolong the life tool so reaming is applied for fitting holes and taper holes.

Design and types of reamer :

There are two types of reamers one is hand reamers and another one is machine reamers. Both are mainly differing with respect to the shank and to the cutting portion.



HAND REAMER



MACHINE REAMER

Hand reamers are used primarily for assembly work to improve the fitting of parts. A long taper lead is the main features of this reamers. This ensures that the hole is well guided and prevents canting. Through the milled square and a tap wrench with clockwise rotation and slight pressure, the reamers are inserted into the hole.

Machine reamers are used for reaming on drilling machines and lathe machines. The work spindle guides them. Lead, therefore, does not need to be as long as hand reamers. Cutting is by the lead with reaming, while the straight portion is used to smooth the hole.

Applications of the reaming process:

Any components requiring precise, cylindrical or tapered holes with good surface finish, either blind or through, such as drilling, after primary hole-making.

Advantages of the reaming process:

- The rates of production range from 10-500/h.
- Low cost of tooling and equipment.
- Finishing costs low and also required cleaning and deburring as well.
- Decreasing feed rate improves surface finish.
- Most precise holes are centre drilled, drilled, bored and reamed to finished size.

Tap

A tap has threads like a bolt. It has three to four flutes cut across the threads. It can cut threads on the inside of a hole. The flutes on the threads form the cutting edges. It is a multi point cutting tool. It will dig into the walls of the hole as the lower part of the tap is slightly tapered. The shank of the tap is square shaped to enable it to be held by a tap wrench.

Nomenclature of Reamer:

Flutes: Longitudinal channels that provide cutting edges, allow chip removal, and allow cutting fluid to reach the cutting edges.

Flute Length: The length of the flutes not including the cutter sweep.

Land: The section of the reamer between the flutes.

Diameter: The diameter of the reamer, which determines the size of the hole it produces.

Nomenclature of Die (for threading):

Threads: The external threads that are cut on a rod or bolt.

Cutting Edges: The edges that form the threads.

Chasers: Used to clean up and refine the threads.

Nomenclature of Taps (for threading):

Threads: The internal threads that are cut in a hole.

Flutes: Longitudinal grooves that allow chips to escape and coolant to flow.

Chamfer Length: The length of the tapered section at the tip of the tap.

Drilling machine operations

1. Drilling
2. Reaming
3. Boring
4. Counter boring
5. Countersinking
6. Spot facing
7. Tapping
8. Trepanning

1.Drilling

A cutting process that involves spinning a drill bit to create a circular hole in solid materials is known as drilling. The drill bit is typically a multi-point rotary cutting tool.

2.REAMING

Reaming is a finishing process that is performed with the multi-edged tools that provide high precision holes and also used for enlarging or finishing a hole previously drilled without changing the chemical and physical properties.

3.Boring

A cutting process that entails enlarging an existing hole in a workpiece with a single-point cutting tool or boring head is known as boring. Sometimes, the process is done to correct the roundness of the previously drilled hole.

4.Counterboring

The process of enlarging a hole over a particular section in the workpiece for screw clearance using counterboring tools is known as counterboring. This creates a shallow and enlarged cylindrical cut in the upper end of the hole to make a place for the head of a screw.

5.Countersinking

The process of enlarging one end of the hole giving it a conical shape is called countersinking. Countersinking creates a V-shaped edge near the hole's surface. It is frequently employed for deburring drilled or tapped holes, and to ensure fasteners sit tightly in the holes.

6.Spot facing

A technique used to create a flat edge on a cylindrical workpiece is called spot facing. This operation involves smoothing and squaring the top surface of the hole for a nut or screw head seat. This is also possible with end-mill cutters.

7.Tapping

The process of creating internal threads using a tool known as the tap is referred to as tapping. This allows a cap screw or bolt to be threaded into the hole after it has been tapped.

8.Lapping

A type of machining in which two surfaces are rubbed against one another with an abrasive between them is Lapping. The lapping operation can be done on these machines where the main outcome of the lapping procedure is to produce truly flat and smooth surfaces.

9.Grinding

A machining operation that uses a grinding wheel to remove the material from a workpiece is grinding. To finish or improve the roundness of a hole that has been drilled, the machine can perform a grinding operation.

10.Trepanning

The process of creating a hole by removing metal using a hollow cutting tool is trepanning. This procedure is used to create big or larger diameter holes. While the hole is being created, fewer chips are removed and a large amount of material is conserved.

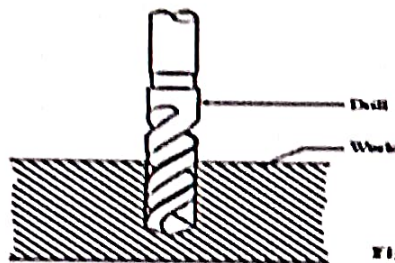


Fig 2.17 Drilling

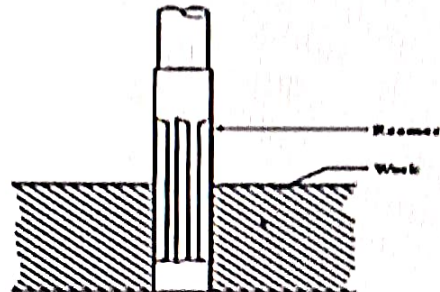


Fig 2.18 Reaming

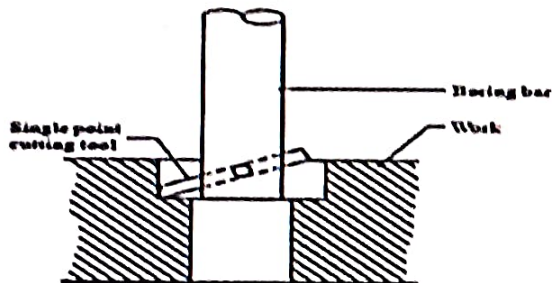


Fig 2.19 Boring

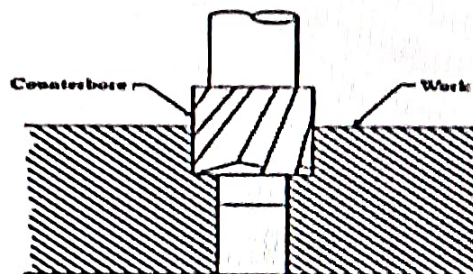


Fig 2.20 Counterboring

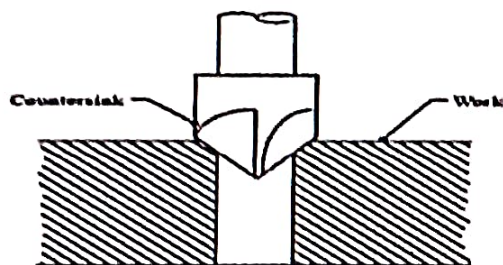


Fig 2.21 Countersinking

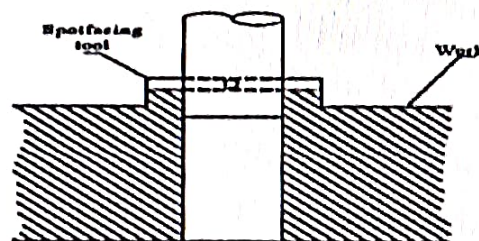


Fig 2.22 Spotfacing

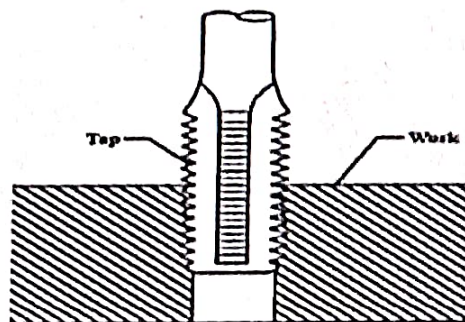


Fig 2.23 Tapping

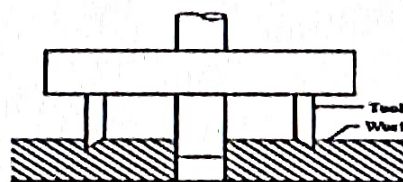
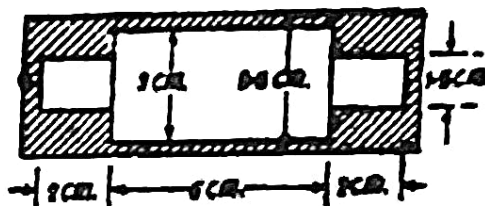


Fig 2.24 Trepanning

Example 32. Estimate the manufacturing cost of 100 pieces of a component shown in Fig. 6'18. These components are to be made from a 3.5 cm. mild steel stock. Assume the following data :

- (i) Density of material = 8 gr./c.c.
- (ii) Material cost = Rs. 2/kg.
- (iii) Parting off and facing allowance/piece = 1 cm.
- (iv) Cost of machining = 2 paise/cu cm. of material removed
- (v) Overhead charges = 50% of labour cost.



Solution. Material Cost

Length of stock required/piece

$$= 6 + 2 + 2 + 1$$

$$= 11 \text{ cm. (considering 1 cm. as parting allowance)}$$

∴ Volume of material required/piece

$$= \frac{\pi}{4} (3.5)^2 \times 11 = 106 \text{ cm.}^3$$

∴ Wt. of material/piece = 106×0.008

$$= 0.848 \text{ kg.}$$

∴ Material cost/piece

$$= 0.848 \times 2 = \text{Rs. } 1.70.$$

Machining Cost :

Volume of material removed

$$= \text{Volume of bar stock} - \text{volume of finished component}$$

$$= 106 - \left\{ \frac{\pi}{4} (1.5)^2 \times 4 + \frac{\pi}{4} (3)^2 \times 6 \right\}$$

$$= 106 - \frac{\pi}{4} (9 + 54) = 56.6 \text{ cu. cm.}$$

∴ Machining cost @ Re. 0.02 cu. cm. of material removed

$$= 56.6 \times 0.02 = \text{Rs. } 1.13$$

Overhead charges = 50% of labour cost

$$= 0.5 \times 1.13 = \text{Re. } 0.57.$$

∴ Manufacturing cost/piece

$$= \text{Material cost} + \text{Machining cost} + \text{Overhead charges}$$

$$= 1.70 + 1.13 + 0.57$$

$$= \text{Rs. } 3.40$$

∴ Cost of 100 pieces = 3.40×100

$$= \text{Rs. } 340. \text{ Ans.}$$